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KTX 887

Due: 9.22.26

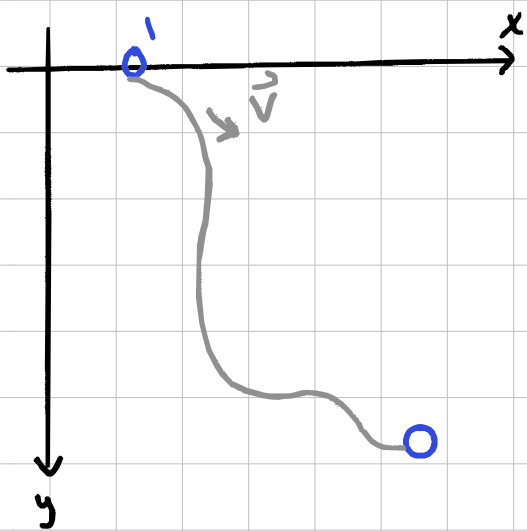
Pset: ch 2 {1, 3, 5, 13, 14, 18}

CM Phy 5103 - fall '26

2.1 Complete Sol. to brachistocrane from Section 2.2

↳ show its a cycloid w/ cusp at x_i where particle is released.

↳ show if Particle is projected w/ 0.g KE $\frac{1}{2} m v_0^2$ that it is still a cycloid but passing through 2 points at z height above y_0 . $\parallel v_0^2 = 2gz$



One of my favorite Physics lore Problems: "know lion by its claw"

I'll note immediately:

- Integrand has no explicit x dependence
- given particle initial Speed v_0 is just vertical translation: $y \mapsto y + z$
- Let $y +$ be downward from release.

$$\text{COE: } \frac{1}{2} m v^2 = mg(+y)$$

$$\hookrightarrow v^2 = 2gy$$

$$v = \sqrt{2gy}$$

So an arc Length Element

$$ds = \sqrt{dx^2 + dy^2} \quad \text{and} \quad dy = y' dx$$

$$ds = \sqrt{dx^2 + y'^2 dx^2} = \sqrt{1 + y'^2} \quad \dots (1)$$

2.1 cont:

So the travel time is: $T[y] = \int \frac{ds}{v} = \int \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx$

and $T[y] = \int f(y, y') dx$, $f(y, y') = \sqrt{\frac{1+y'^2}{2gy}} = \frac{\sqrt{1+y'^2}}{\sqrt{2gy}}$

Clearly No x dependence: $\left(\frac{\partial f}{\partial x} = 0\right)$ then Beltrami $\rightarrow$

$$f - y' \frac{\partial f}{\partial y'} = c.$$

$$\frac{\partial f}{\partial y'} = \frac{1}{\sqrt{2gy}} \frac{\partial}{\partial y'} (1+y'^2)^{1/2} = \frac{1}{\sqrt{2gy}} \left[\frac{1}{2} (1+y'^2)^{-1/2} \cdot 2y' \right]$$

$$\boxed{\frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{2gy} \cdot \sqrt{1+y'^2}}} \dots (2)$$

$$\text{then } f - y' \frac{\partial f}{\partial y'} = \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} - \frac{y'}{\sqrt{2gy} \cdot \sqrt{1+y'^2}}$$

$$\hookrightarrow f - y' \frac{\partial f}{\partial y'} = \frac{1}{\sqrt{2gy}} \left(\sqrt{1+y'^2} - \frac{y'^2}{\sqrt{1+y'^2}} \right)$$

$$= \frac{1}{\sqrt{2gy} \cdot \sqrt{1+y'^2}} = \boxed{\frac{1}{\sqrt{2gy(1+y'^2)}} = c}$$

Choose $C = \frac{1}{\sqrt{4ga}} \rightarrow \text{then: } \frac{1}{\sqrt{2gy \cdot (1+y'^2)}} = \frac{1}{\sqrt{4ga}}$

$$\downarrow \frac{1}{2gy(1+y'^2)} = \frac{1}{4ga}$$

$$\downarrow 2gy(1+y'^2) = 4ga$$

$$\boxed{y(1+y'^2) = 2a}$$

2.1 cont:

$$y'^2 = \frac{2a}{y} - 1 = \frac{2a-y}{y}; \quad \text{then for desc brace} \quad \frac{dy}{dx} = \sqrt{\frac{2a-y}{y}}$$

$$\text{Let } y = a(1 - \cos \theta)$$

$$\hookrightarrow 2a - y = 2a - a(1 - \cos \theta) = a(1 + \cos \theta)$$

$$\hookrightarrow \left(\frac{dy}{dx}\right)^2 = \frac{1 + \cos \theta}{1 - \cos \theta}$$

$$\left[\begin{array}{l} \text{Identities: (half angle?)} \\ 1 + \cos \theta = 2 \cos^2(\theta/2) \\ 1 - \cos \theta = 2 \sin^2(\theta/2) \end{array} \right]$$

$$\frac{dy}{dx} = \cot(\theta/2) \quad \text{and} \quad \frac{dx}{dy} = \tan(\theta/2)$$

$$[\sin \theta = 2 \sin(\theta/2) \cos(\theta/2); \sin \theta \tan \theta/2 = 2 \sin(\theta/2) \cos(\theta/2) \cdot \frac{\sin \theta/2}{\cos \theta/2}]$$

$$dx = a(1 - \cos \theta) d\theta$$

$$x = a \cdot \int (1 - \cos \theta) d\theta = a(\theta - \sin \theta) + C_1$$

$$\hookrightarrow \text{at release: } \theta = 0; \quad x = 0 \therefore C_1 = 0$$

thus

$$x = a(\theta - \sin \theta), \quad y = a(1 - \cos \theta) \quad \text{cycloid}$$

2.1 cont: Cusp at initial point

diff parametric: $\frac{dx}{d\theta} = a(1 - \cos\theta)$; $\frac{dy}{d\theta} = a \sin\theta$

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sin\theta}{1 - \cos\theta}$$

$$[1 - \cos\theta = 2\sin^2\frac{\theta}{2}]$$

$$= \frac{2 \cdot \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right)}{2\sin^2\left(\frac{\theta}{2}\right)} = \cot\left(\frac{\theta}{2}\right)$$

so: $\frac{dy}{dx} \rightarrow \infty$; as $\theta \rightarrow 0^+$; for local, I'll expand near $\theta = 0$

$$\sin\theta = \theta - \frac{\theta^3}{6} + O(\theta^5)$$

$$\hookrightarrow x = a - (\theta \cdot \sin\theta) = a \left[\theta - \left(\theta - \frac{\theta^3}{6} + O(\theta^5) \right) \right]$$

$$\text{then } x = \frac{a\theta^3}{6} + O(\theta^5) \quad \text{and} \quad \cos\theta = 1 - \frac{\theta^2}{2} + O(\theta^4)$$

$$\hookrightarrow y = \frac{a\theta^2}{2} + O(\theta^4)$$

to lowest order:

$$x \approx \frac{a\theta^3}{6} ; \quad y \approx \frac{a\theta^2}{2}$$

2.1 cont:

$$y \approx \frac{a\theta^2}{2} \rightarrow \theta^2 \approx \frac{2y}{a} \rightarrow \theta^6 \approx \frac{8y^3}{a^3}$$

$$\text{Similarly: } x^2 \approx \frac{a^2\theta^6}{36} \approx \frac{a^2}{36} \cdot \frac{8y^3}{a^3} \approx \frac{2y^3}{9a}$$

$$y^3 \approx \frac{9a}{2} x^2$$

and if particle non zero $v_0 \neq 0$ CoE: $\frac{1}{2}mv^2 = \frac{1}{2}mv_0^2 + mgy$

$$v_0^2 = 2g \cdot z$$

$$\text{then } v^2 = 2g(y+z) \rightarrow v = \sqrt{2g(y+z)}$$

$$v = \sqrt{2g(y+z)}$$

$$(y = b+z)$$

$$T[y] = \int \frac{\sqrt{1+y'^2}}{\sqrt{2g(y+z)}} dx = \int \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx \quad \star \text{ same functional as } v_0=0$$

$$\therefore x = x_c + a(\theta - \sin\theta) \rightarrow y = a(1 - \cos\theta) - z$$

$$\text{at cusp } \theta=0 \rightarrow y=0 \rightarrow y+z=0 \therefore y=-z$$

+y is downward; $y=-z$ is a point vertical distance z

$$\text{above physical starting point } \left(z = \frac{v_0^2}{2g} \right)$$

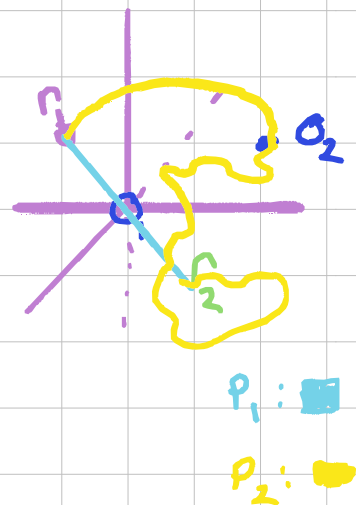
P. 2.3

Prove the shortest distance between two points in space is a straight line:

Let a path which joins $r_1 + r_2$ (2 points in space) be:

$$r(\lambda) = x(\lambda) \hat{x} + y(\lambda) \hat{y} + z(\lambda) \hat{z}$$

$$ds = \sqrt{dx^2 + dy^2 + dz^2}$$



Since: $dx = \frac{\partial x}{\partial \lambda} d\lambda$ [similarity for dy & dz]

then:

$$ds = \sqrt{\left(\frac{\partial x}{\partial \lambda}\right)^2 + \left(\frac{\partial y}{\partial \lambda}\right)^2 + \left(\frac{\partial z}{\partial \lambda}\right)^2} d\lambda \quad \text{and thus the}$$

total length of the functional is:

$$L[r] = \int_{\lambda_1}^{\lambda_2} \sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2} d\lambda \quad (\text{dot means } d/d\lambda)$$

Apply E-L: but f has no explicit dependence on x

$$\therefore \frac{\partial f}{\partial x} = 0 \rightarrow \frac{d}{d\lambda} \left(\frac{\partial f}{\partial \dot{x}} \right) = 0$$

$$\text{but } \frac{\partial f}{\partial \dot{x}} = \frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}} \rightarrow \frac{d}{d\lambda} \left[\frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}} \right] = 0$$

Similarly

$$\frac{d}{d\lambda} \left[\frac{\dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}} \right] = 0$$

$$\frac{d}{d\lambda} \left[\frac{\dot{z}}{\sqrt{\dot{x}^2 + \dot{y}^2 + \dot{z}^2}} \right] = 0$$

2.3 cont

So in vector notation $\frac{d}{d\lambda} \left(\frac{\dot{\vec{r}}}{|\dot{\vec{r}}|} \right) = 0$

which is simply the change of unit tangent vector w/ respect to λ .

The unit tangent to the path is constant ($\hat{t} = \frac{\dot{\vec{r}}}{|\dot{\vec{r}}|} = \text{const.}$)

tangent never changes $\therefore$ path has zero curvature and

is a straight line:

P: 2.5: a particle in potential: $V(x) = -Fx$, F is const.

Particle travels from $x=0$ to $x=a$ in time t_0 .

find A, B, C || E.O.M. are expressed $x(t) = A + Bt + Ct^2$

$$V(x) = -Fx \quad ; \quad x(0) = 0; \quad x(t_0) = a$$

We want A, B, C such that the action is minimized.

at $t=0$

$$- x(0) = A = 0;$$

therefore $A=0$ at $t=t_0$

$$\hookrightarrow x(t_0) = Bt_0 + Ct_0^2 = a$$

$$\text{thus } Bt_0 = a - Ct_0^2 \rightarrow B = \frac{a}{t_0} - Ct_0$$

★ this is reduced and can be written as a family of paths w/ only C .

$$x(t) = \left(\frac{a}{t_0} - Ct_0 \right) t + Ct^2$$

$$\mathcal{L} = \left(\frac{1}{2} m \dot{x}^2 \right) + (F \cdot x)$$

$$\dot{x} = B + 2Ct \quad \therefore \mathcal{L} = \frac{1}{2} m (B + 2Ct)^2 + F(Bt + Ct^2)$$

$$\text{action: } S = \int_0^{t_0} \left(\frac{1}{2} m (B + 2Ct)^2 + F(Bt + Ct^2) \right) dt$$

$$= \int_0^{t_0} \left[\frac{1}{2} m B^2 + 2mB Ct + 2mC^2 t^2 + FBt + FCt^2 \right] dt$$

2.5 cont.:

$$\hookrightarrow B = \frac{a}{t_0} - Ct_0$$

Integrating term by term:

$$\int_0^{t_0} \frac{1}{2} m B^2 dt + \int_0^{t_0} (2m B(t)) dt + \int_0^{t_0} 2mc^2 t^2 dt + \int_0^{t_0} FBt dt + \int_0^{t_0} F \cdot C \cdot t dt$$

$$S(C) = \frac{ma^2}{2t_0} + \frac{Fat_0}{2} + \frac{mC^2 t_0^3}{6} - \frac{F \cdot C t_0^3}{6}$$

Since first 2-terms do not depend on C , the action that minimizes requires:

$$\frac{\partial S}{\partial C} = 0 = \left(\frac{mCt_0^3}{3} - \frac{Ft_0^3}{6} \right) \quad \text{and } t_0 \neq 0$$

thus

$$\frac{mC}{3} = \frac{F}{6} \rightarrow \boxed{C = \frac{F}{2m}} \dots (2)$$

$$\boxed{A = 0}$$

$$B = \frac{a}{t_0} - \left(\frac{F}{2m} \right) t$$

$$\boxed{B = \frac{a}{t_0} - \frac{Ft_0}{2m}}$$

thus: minimizing trajectory

$$\boxed{\chi(t) = \left(\frac{a}{t_0} - \frac{Ft_0}{2m} \right) t + \frac{F}{2m} t^2}$$

Sanity check: $\frac{\partial^2 S}{\partial C^2} = \frac{m t_0^3}{3}$ for $m > 0$; $t_0 > 0$

✓ minimum within quadratic fam.

$$\ddot{\chi} = 2C \quad ; \quad C = F/2m, \Rightarrow m\ddot{\chi} = F \quad \checkmark$$

We have independently recovered constant acceleration

trajectory w/ variational minimum.

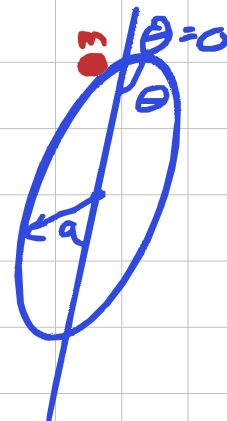
2.13

Heavy particle at top of verticle hoop.

use lagrange + lagrange undetermined multipliers to find height that particle falls off.

• Assuming particle is frictionless; placed up to; unstable equilibrium:

↳ constraint imposed: $f(r) = r - a = 0$



$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$$

Verticle coordinate $y = r \cos \theta \rightarrow V = mgr \cdot \cos \theta$

$$\mathcal{L} = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) - mgr \cos \theta \quad \dots (1)$$

This is a holonomic constraint:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = \lambda \frac{\partial f}{\partial q_i} \quad ; \quad \text{bc } f = r - a$$

then $\frac{\partial f}{\partial r} = 1$; $\frac{\partial f}{\partial \theta} = 0$; therefore the multiplier λ acts entirely in the radial direction (and is reaction force of the hoop)

$$r: \frac{\partial \mathcal{L}}{\partial r} = m\ddot{r} \quad \text{so:} \quad \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) = m\ddot{r}$$

$$\text{and } \frac{\partial \mathcal{L}}{\partial r} = mr\dot{\theta}^2 - mg \cos \theta$$

$$\hookrightarrow \boxed{m\ddot{r} - mr\dot{\theta}^2 + mg \cdot \cos \theta = \lambda} \quad \dots (2)$$

2.13 cont.

Our hoop has constraints: $r=a \rightarrow \dot{r} = \ddot{r} = 0$

thus $\lambda = mg \cdot \cos \theta - ma \dot{\theta}^2$; constrained force is radial

then $N = \lambda = mg \cos \theta - ma \dot{\theta}^2$, equivalently: $v = a \dot{\theta}$

then: $N = mg \cdot \cos \theta - \frac{mv^2}{a}$ hoop reaction while contact is maintained.

bc frictionless $\rightarrow$ energy conservation

Initially: $\theta = 0, v = 0$

$E_i = m \cdot g \cdot a$; at a given angle θ

$$E = \frac{1}{2}mv^2 + (mga \cdot \cos \theta)$$

$$\hookrightarrow \frac{1}{2}mv^2 + mga \cos \theta = mga$$

$$\hookrightarrow \frac{1}{2}v^2 = g \cdot a (1 - \cos \theta) \rightarrow v^2 = 2ga(1 - \cos \theta)$$

and $v = a \dot{\theta}$

$$a^2 \dot{\theta}^2 = 2ga \cdot (1 - \cos \theta) \rightarrow a \dot{\theta}^2 = 2g(1 - \cos \theta)$$

Reaction as funct. position



$$\text{Recall: } N = mg \cos \theta - ma \dot{\theta}^2 = mg \cos \theta - 2mg(1 - \cos \theta)$$

$$= mg \cdot \cos \theta - 2mg + 2mg \cdot \cos \theta$$

$$N = m \cdot g (3 \cos \theta - 2)$$

height of fall off

$N = m \cdot g (3 \cos \theta - 2)$; particle loses contact when Normal reaction is zero.

$$\therefore 3 \cdot \cos \theta - 2 = 0$$

$$\hookrightarrow \cos \theta = 2/3 \quad (y = a \cos \theta) \rightarrow$$

$$y = \frac{2 \cdot a}{3} \quad \text{above center}$$

or

$$y = \frac{a}{3} \quad \text{from top}$$

Problem 2.14

Same as P: 2.13 + rot KE + rolling constraint

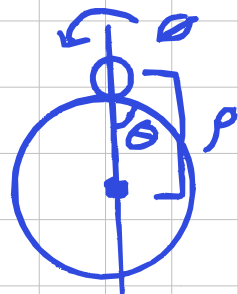
Uniform hoop of mass m and radius r : Starts at rest up top
and use Lagrange multipliers

Let $a \equiv R+r$ while hoop remains in contact,
center of small hoop moves on a circle of radius a .

Take: ρ = distance between centers

θ = angle from vertical upward

ϕ = rotation angle of small hoop



$$q = (\rho, \theta, \phi)$$

i) $f_1 = \rho - a = 0$: constraint while hoop remains on cylinder

ii) $a \cdot d\theta$: rolling w/o slipping; center tracks in arc.

↓ small hoop

$$r d\phi \rightarrow \therefore a \cdot d\theta = r d\phi$$

zero constraint $\rightarrow f_2 = r\phi - a\theta = 0$

$$\therefore \underline{r\dot{\phi} = a\dot{\theta} \text{ and } r\ddot{\phi} = a\ddot{\theta}}$$

3 coordinates - 2 constraints = 1 DoF

$$V_c^2 = \dot{\rho}^2 + \rho^2 \dot{\theta}^2 \quad \text{polar speed:}$$

$$T_{\text{trans}} = \frac{1}{2} m (\dot{\rho}^2 + \rho^2 \dot{\theta}^2) \quad \text{Translational KE}$$

$$T = \frac{1}{2} I \dot{\phi}^2 \rightarrow I = mr^2 \quad \text{hoop}$$

$$\therefore T = \frac{1}{2} m (\dot{\rho}^2 + \rho^2 \dot{\theta}^2) + \frac{1}{2} mr^2 \dot{\phi}^2 \quad (\text{fixed cylinder center } y=0)$$

$$y_c = \rho \cdot \cos \theta \text{ then } V = mg\rho \cos \theta$$

$$\therefore \mathcal{L} = \frac{1}{2} m (\dot{\rho}^2 + \rho^2 \dot{\theta}^2) + \frac{1}{2} mr^2 \dot{\phi}^2 - mg\rho \cos \theta$$

2.14 cont:
multipliers

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \lambda \frac{\partial f_1}{\partial q_i} + \mu \frac{\partial f_2}{\partial q_i}$$

we already defined $f_1 = \rho - a$ and $f_2 = r\rho - a\theta \rightarrow \lambda$ will be normal force

$$\frac{\partial \mathcal{L}}{\partial \dot{\rho}} = m\dot{\rho} \rightarrow \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\rho}} = m\ddot{\rho} \quad \text{and} \quad \frac{\partial \mathcal{L}}{\partial \rho} = m\rho\dot{\theta}^2 - mg \cos \theta$$

$$\frac{\partial f_1}{\partial \rho} = 1, \quad \frac{\partial f_2}{\partial \rho} = 0 \rightarrow m\ddot{\rho} - m\rho\dot{\theta}^2 + mg \cdot \cos \theta = \lambda$$

$\hookrightarrow$ apply contact restraint ($\rho = a, \dot{\rho} = \ddot{\rho} = 0$)

$$N = mg \cdot \cos \theta - ma\dot{\theta}^2 \quad \leftarrow \text{where hoop falls off}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m\rho^2\dot{\theta} \rightarrow \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) = m\rho^2\ddot{\theta} + 2m\rho\dot{\rho}\dot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = mg\rho \cdot \sin \theta \quad \text{and we know} \quad \frac{\partial f_1}{\partial \theta} = 0, \quad \frac{\partial f_2}{\partial \theta} = -a$$

$$m\rho^2\ddot{\theta} + 2m\rho\dot{\rho}\dot{\theta} - mg\rho \sin \theta = -a\mu$$

$$\rightarrow ma^2\ddot{\theta} - mga \sin \theta = -a\mu \quad \dots (1)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mr^2\dot{\theta} \rightarrow \frac{d}{dt} (mr^2\dot{\theta}) = mr^2\ddot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = 0 \quad \text{and} \quad \frac{\partial f_2}{\partial \theta} = r \rightarrow mr^2\ddot{\theta} = \mu r$$

$$\text{So } \mu = mr\ddot{\theta} \xrightarrow{\text{rolling const.}}$$

$$\mu = ma\ddot{\theta} \quad \dots (2)$$

eq 2 $\rightarrow$ eq 1

$$\hookrightarrow ma^2\ddot{\theta} - mga \sin \theta = -a(ma\ddot{\theta})$$

$$\hookrightarrow ma^2\ddot{\theta} + ma^2\ddot{\theta} = mga \sin \theta \rightarrow 2ma^2\ddot{\theta} = mga \sin \theta$$

$$\ddot{\theta} = \frac{g}{2a} \sin \theta = \frac{g}{2(R+r)} \cdot \sin \theta = \ddot{\theta}$$

2.13 cont.:

$\dot{\theta}(\theta)$ is goal: start $\theta=0$, $\dot{\theta}>0$, coE rather than $\int$

$$\Delta v = mga(1 - \cos \theta), \quad T = ma^2 \dot{\theta}^2$$

$$\hookrightarrow ma^2 \dot{\theta}^2 = mga(1 - \cos \theta)$$

$$\hookrightarrow a \dot{\theta}^2 = g(1 - \cos \theta)$$

Contact only possible while $N \geq 0$

$$N = mg \cos \theta - ma \dot{\theta}^2$$

$$\hookrightarrow mg \cdot \cos \theta = ma \dot{\theta}^2 \rightarrow g \cos \theta = a \dot{\theta}^2 \quad \downarrow \text{eq (3)}$$

$$\hookrightarrow g \cdot \cos \theta = g(1 - \cos \theta)$$

$$\hookrightarrow \cos \theta = 1/2 \rightarrow \theta = \pi/3 = 60^\circ \text{ loop loses contact}$$

2.18

one generalized coordinate from bottom of hoop.

Let θ be particle angle from downward vertical. Hoop

rotates about its vertical $\rightarrow$ constant angular speed ω

$$\phi = \omega t$$

in cylindrical for Particle: $\rho = a \cdot \sin \theta$; $z = -a \cos \theta$

$$v^2 = \dot{\rho}^2 + \rho^2 \dot{\theta}^2 + \dot{z}^2 \quad \dots (1)$$

$$\dot{\rho} = a \cos \theta \cdot \dot{\theta}, \quad \dot{z} = a \sin \theta \cdot \dot{\theta}, \quad \dot{\phi} = \omega$$

$$\hookrightarrow v^2 = a^2 \dot{\theta}^2 + a^2 \omega^2 \sin^2 \theta$$

$$\therefore T = \frac{1}{2} m a^2 (\dot{\theta}^2 + \omega^2 \sin^2 \theta) \quad \dots (2)$$

$$\text{Potential: } V = mgz = -mga \cdot \cos \theta$$

$$\hookrightarrow \mathcal{L} = \frac{1}{2} m a^2 \dot{\theta}^2 + \frac{1}{2} m a^2 \omega^2 \sin^2 \theta + m g a \cos \theta$$

$$\hookrightarrow \frac{\partial}{\partial t} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\text{i) } \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m a^2 \dot{\theta} \rightarrow \frac{d}{dt} (m a^2 \dot{\theta}) = m a^2 \ddot{\theta}$$

$$\text{ii) } \frac{\partial \mathcal{L}}{\partial \theta} = m a^2 \omega^2 \sin \theta \cos \theta - m g a \sin \theta$$

$$\therefore m a^2 \ddot{\theta} - m a^2 \omega^2 \sin \theta \cos \theta + m g a \sin \theta = 0$$

$$\hookrightarrow \ddot{\theta} = \sin \theta (\omega^2 \cos \theta - g/a)$$

Since

$\mathcal{L}$ has no dependence explicitly on time

$$h = \dot{\theta} \cdot \frac{\partial \mathcal{L}}{\partial \dot{\theta}} - \mathcal{L} \quad \text{is constant}$$

$$\hookrightarrow h = \frac{1}{2} m a^2 \dot{\theta}^2 - \frac{1}{2} m a^2 \omega^2 \sin^2 \theta - m g a \cos \theta = \text{constant}$$

and Stationary solutions: for particle to remain

$$\dot{\theta} = \ddot{\theta} = 0$$

$$\text{hence: } \sin \theta (\omega^2 \cos \theta - g/a) = 0$$

$$\text{bottom angle } \theta = 0$$

$$\text{non bottom angle requires } \omega^2 \cos \theta = g/a \quad \swarrow \sin(\theta)$$

$$\hookrightarrow \cos \theta = g/a\omega^2$$

$$\hookrightarrow \frac{g}{a\omega^2} \leq 1 \quad \text{therefore } \omega^2 \geq g/a$$

$$\omega_0 = \sqrt{\frac{g}{a}} \rightarrow \omega < \omega_0 : \text{non bottom}$$

and for stationary points away from bottom

$$\omega > \omega_0 : \theta_0 = \cos^{-1} \left(\frac{g}{a\omega^2} \right)$$